Virtual Private Network



Sub-Topics:

Introduction to VPN, VPN Architecture, Types of VPN, VPN tunneling, VPN protocol (PPTP, L2TP, SSTP, OpenVPN, WireGuard), Applications of VPN

A **Virtual Private Network (VPN)** is a technology that creates a **secure and encrypted connection** over a public network (like the internet). It allows users to send and receive data **as if they are directly connected to a private network**.

Normally, when you access the internet:

- Your data is **visible** to ISPs, hackers, or attackers (especially on public Wi-Fi)

With VPN:

- Your data is **encrypted**
- Your **IP address is hidden**
- Your online activity becomes **private and secure**




VPN
with VPN


without VPN

How VPN Works

- User connects to a VPN client
- VPN client encrypts the data
- Data travels through a secure “tunnel”
- VPN server decrypts the data and sends it to the internet
- Response comes back through the same secure tunnel

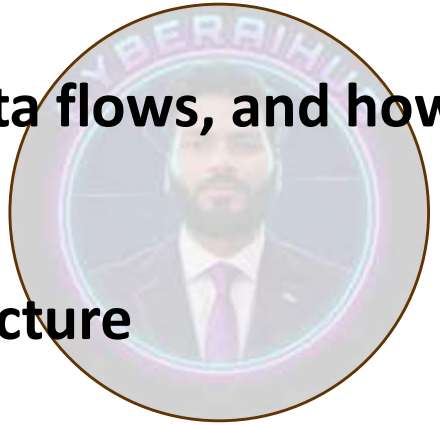


🔐 VPN Architecture

📌 What is VPN Architecture?

VPN Architecture defines how different components (devices, servers, protocols) are structured and interact to provide a **secure communication tunnel** over the internet.

👉 It explains **who connects, how data flows, and how security is enforced**.



🔑 Core Components of VPN Architecture

1. VPN Client

- Installed on user devices (laptop, mobile)
- Initiates VPN connection
- Encrypts and decrypts data

2. VPN Server (Gateway)

- Receives encrypted data from clients
- Decrypts and forwards it to the internet or private network
- Acts as a **bridge between user and destination**

3. Authentication Server

- Verifies user identity (username/password, certificates, MFA)
- Ensures only authorized users access the VPN



4. Tunnel (Secure Channel)

- Logical encrypted path between client and server
- Uses protocols like IPSec, SSL/TLS, OpenVPN

5. Encryption Mechanism

- Protects confidentiality of data
- Common algorithms: AES, RSA



Types of VPN

1. Remote Access VPN Architecture

☞ Used by individual users (most common)

Flow:

- User → VPN Client → Internet → VPN Server → Private Network

Example:

- Employee working from home accessing company resources

Key Features:

- Secure login (authentication)
- Encrypted communication
- Flexible (works from anywhere)



Remote Access VPN Architecture



Secure Access

Data Encryption

User Authentication

2. Site-to-Site VPN Architecture

☞ Used between organizations or branches

Flow:

- Branch Office Router → Internet → Head Office VPN Gateway

Types:

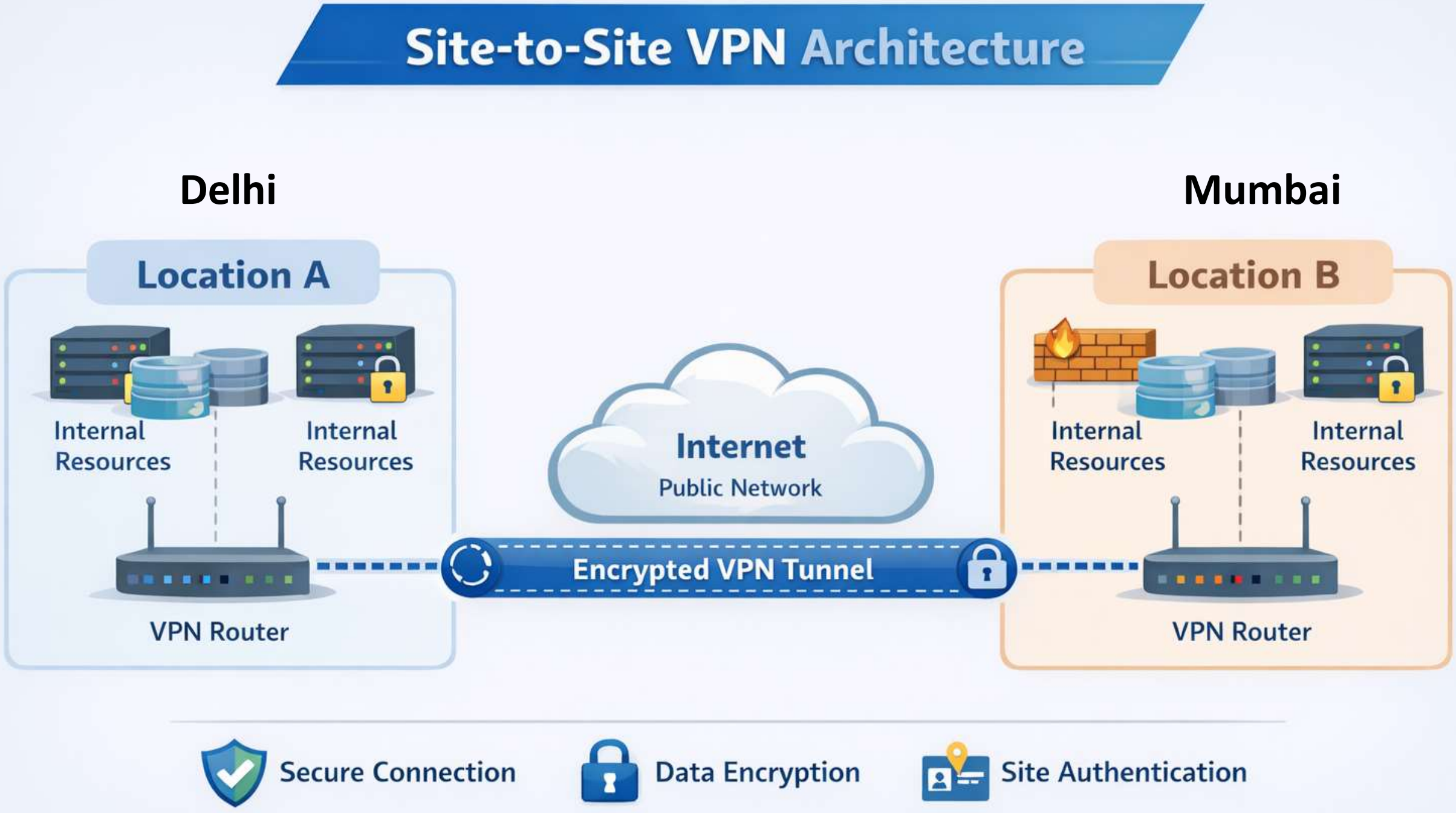
- **Intranet VPN** → Connects branches of same company
- **Extranet VPN** → Connects with partners/suppliers



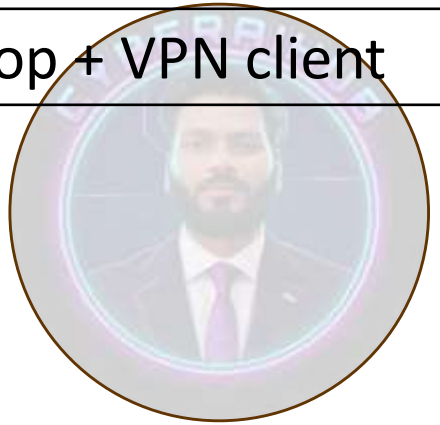
Key Features:

- No need for individual client software
- Always-on connection
- Used in enterprise environments

Site-to-Site VPN Architecture



Feature	Remote Access VPN	Site-to-Site VPN
Users	Individual users	Entire networks
Login	Required	Not required (auto)
Example	Work from home	Bank branches
Device	Laptop + VPN client	VPN routers/firewalls



🔒 VPN Tunneling

VPN Tunneling is the process of creating a **secure, encrypted path (tunnel)** between a user's device and a VPN server over the internet.

👉 It allows data to travel safely through an **untrusted network (like the internet)** without being exposed.

- 🎁 Your data = a package
- 🔒 Encryption = Locked box
- 🚇 Tunnel = secure underground path

Even if someone intercepts the data, they **cannot read it**.

How VPN Tunneling Works

Step-by-Step Flow:

1.User sends data (e.g., website request)

2.VPN client **encrypts** the data

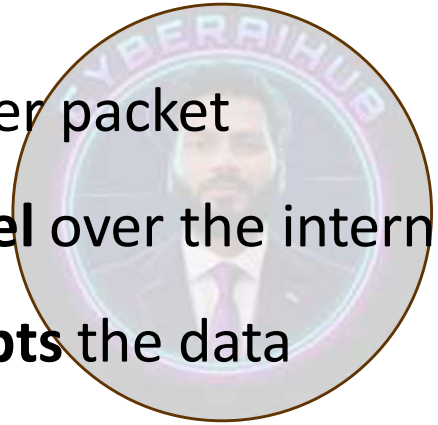
3.Data is **encapsulated** inside another packet

4.It travels through the **secure tunnel** over the internet

5.VPN server **decapsulates & decrypts** the data

6.Data is forwarded to the destination

7.Response comes back through the same tunnel



How VPN Tunneling Works

1 User sends data
(e.g.. website request)



Internet





Encrypted Packet

2 VPN client encrypts
the data



Internet

3 Data is encapsulated
inside another packet



4 VPN server decapsulates
& decrypts the data



Website Server

4 Data is forwarded
through the same tunnel



VPN Server



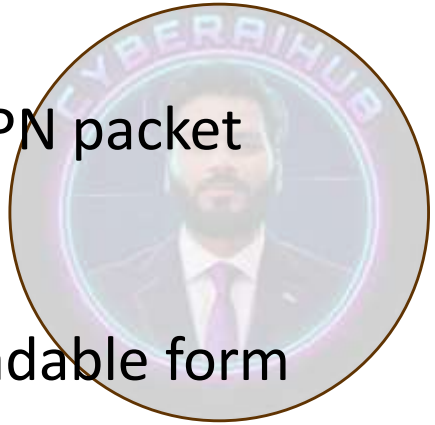
Website

Key Concepts in VPN Tunneling

1. Encapsulation

- Original data packet is wrapped inside another packet
- Adds an extra layer of protection

☞ Example:
Original packet → Wrapped inside VPN packet



2. Encryption

- Converts readable data into unreadable form
- Ensures confidentiality

3. Decapsulation

- Removing the outer packet at the VPN server
- Original data is restored

🌐 Types of VPN Tunneling

1. Split Tunneling
- Only **sensitive traffic** goes through VPN
- Other traffic goes directly to the internet

✅ Faster performance
❌ Less secure

2. Full Tunneling
- **All traffic** goes through VPN

✅ Maximum security
❌ Slightly slower

3. Voluntary Tunneling
- Initiated by the **user/device**
- Most common (e.g., VPN apps)

4. Compulsory Tunneling
- Initiated by **ISP or organization**
- User has no control



Why Tunneling is Important
- Protects data from **hackers/sniffers**
- Ensures **privacy on public Wi-Fi**
- Enables **secure remote access**
- Prevents **data interception (MITM attacks)**

📌 What are VPN Protocols?

VPN Protocols are a set of rules and technologies that determine **how data is encrypted, transmitted, and secured** between a VPN client and a VPN server.

☞ In simple terms:
They decide **how your secure tunnel is created and maintained**.



🔑 Common VPN Protocols

1. IPSec (Internet Protocol Security)

- Works at **Network Layer**
- Uses **AH (Authentication Header)** & **ESP (Encapsulating Security Payload)**
- Often combined with IKE (Key Exchange)

Example:

- Site-to-site VPN between company branches

Purpose:

- Provides **strong encryption & authentication** for enterprise networks

2. OpenVPN

- Open-source protocol using **SSL/TLS**
- Works over **TCP or UDP**

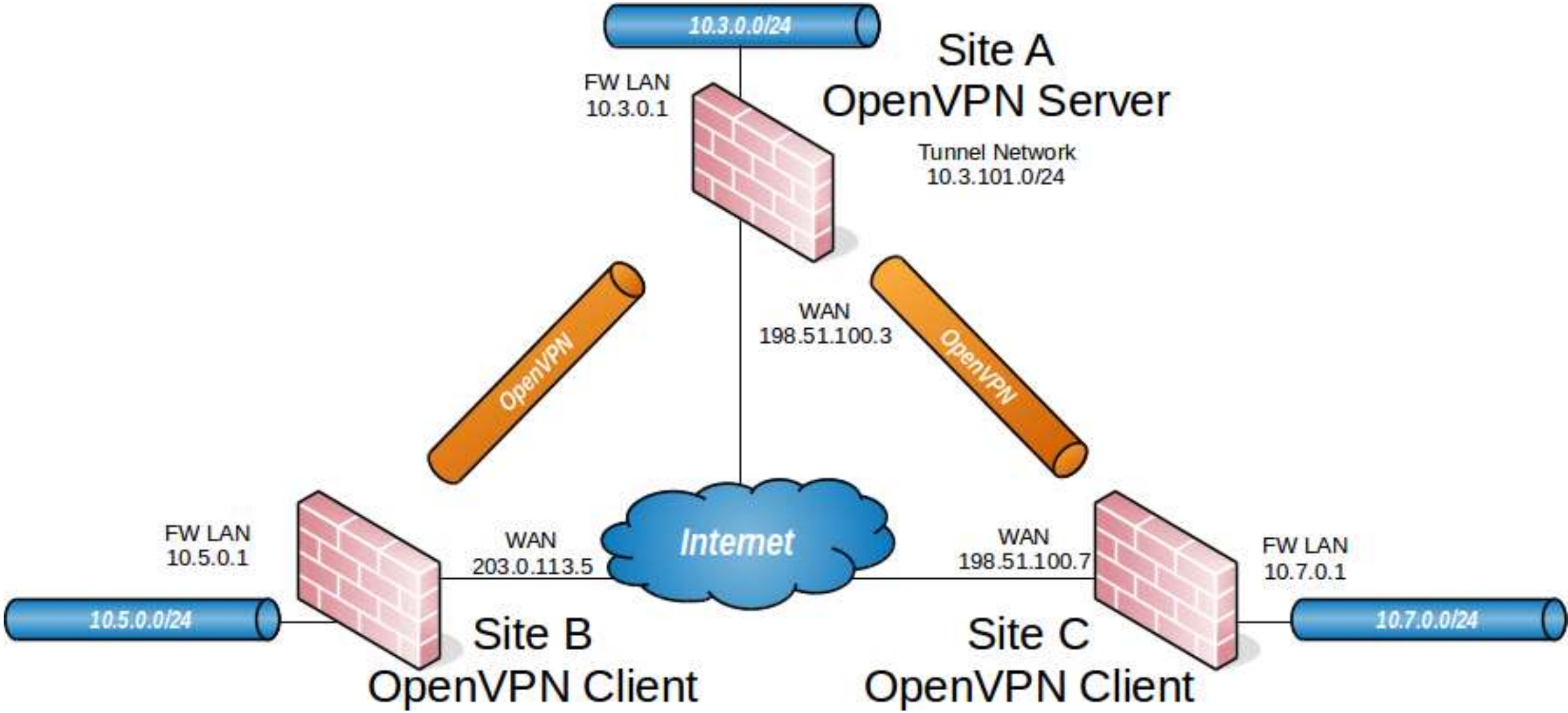
Example:

- VPN apps like enterprise remote access solutions



Purpose:

- Flexible, secure, widely used for **remote access VPNs**


10.3.0.0/24
Site A
OpenVPN Server
FW LAN
10.3.0.1
Tunnel Network
10.3.101.0/24
WAN
198.51.100.3
OpenVPN
OpenVPN
FW LAN
10.5.0.1
WAN
203.0.113.5
Internet
WAN
198.51.100.7
FW LAN
10.7.0.1
10.5.0.0/24
10.7.0.0/24
Site B
OpenVPN Client
Site C
OpenVPN Client

3. WireGuard

- Modern, lightweight protocol
- Uses **state-of-the-art cryptography**

Example:

- High-speed VPN services



Purpose:

- Provides **fast, efficient, and secure communication**

4. L2TP/IPSec (Layer 2 Tunneling Protocol)

Combines **L2TP (tunneling)** + **IPSec (encryption)**

Example:

- Built-in VPN support in Windows

Purpose:

- Provides **secure tunneling with encryption**



PPTP (Point-to-Point Tunneling Protocol)

- One of the oldest VPN protocols
- Uses **GRE tunneling**

Example:

- Legacy systems

Purpose:

- Fast but **less secure (deprecated)**



Comparison of VPN Protocols

Protocol	Security 🔒	Speed ⚡	Use Case	Notes
IPSec	High	Medium	Site-to-Site	Enterprise-grade
OpenVPN	Very High	Medium	Remote Access	Highly flexible
WireGuard	High	Very Fast	Modern VPNs	Lightweight
L2TP/IPSec	High	Medium	Built-in VPN	Double encapsulation
PPTP	Low	Fast	Legacy	Not recommended

💼 Applications of VPN

1. Secure Remote Access

•Employees work from home securely

2. Data Privacy

•IP address hide karta hai

•Tracking reduce karta hai

3. Bypassing Geo-Restrictions

•Blocked websites access kar sakte hain



4. Secure Public Wi-Fi Use

•Coffee shop / airport Wi-Fi par protection

5. Business Communication

•Offices ke beech secure connection

6. Cybersecurity

•Data encryption + safe communication